Climate Opportunity Navigator: MVP Feature Plan

Project Goal (MVP): To validate our core two-sided marketplace.

1. **Innovator (Supply) Side:** Prove we can attract innovators by providing them with high-quality, *unsolved problem statements* from our curated database.
2. **Investor (Demand) Side:** Prove we can create a "deal flow" pipeline of these new, vetted projects that is valuable to VCs and NGOs.

1. Core User System (Shared Backend)

- **Feature:** User Sign-Up & Login
- **Details:**
  - Simple email/password or Google sign-in.
  - Must be a single user system for both Innovators and Investors (VCs).
  - A user's "role" (Innovator vs. Investor) will determine which dashboard they see.
- **Database (Firestore - `users` collection):**
  - `userId`
  - `email`
  - `name`
  - `role` (e.g., "innovator", "investor")
  - `profession` (e.g., "student", "engineer" - from onboarding)
  - `interests` (e.g., ["agri-tech", "water"] - from onboarding)

2. Innovator Journey: Frontend

This is the "AI Discovery Engine" (your free product).

A. New Project Onboarding (The "Chat UI")

- **Trigger:** User clicks "Start a New Project" or "New Chat".
- **Feature:** This is a **one-time form**, not a chat, to get structured data.
- **UI Components:**
  1. `Your Profession:` (MCQ: Student, Engineer, Researcher, Other)
  2. `Your Domain:` (MCQ: Agri-Tech, Water Management, Public Health)
  3. `Your Goal (Optional):` (Text box: "I want to find low-cost hardware solutions...")
- **Action:** User clicks "Discover Problems" -> sends this structured data to the backend.

B. Project Results Page (The "Aha!" Moment)

- **Trigger:** Backend returns a successful response.
- **Feature:** Display the `Top 3 Unsolved Challenges` .
- **UI Components (for each challenge):**

1. **The Challenge:** A 1-2 sentence problem statement (e.g., "A key gap is the lack of low-cost soil moisture sensors...").
2. **The Source:** A direct citation (e.g., "Source: *Nature Sustainability*, 2024").
3. **The SDGs:** Icons for the relevant tags (e.g., `SDG 2` , `SDG 6`).
4. **The "Start Project" Button:** This is the key call-to-action.

C. "My Impact" Page (Innovator Dashboard)

- **Trigger:** User clicks "My Impact" in the main menu.
- **Feature:** A personal gallery of all projects the user has "started".
- **UI Components:**
  - A list of "Project Cards."
  - Each card shows the Problem Statement they are working on.
  - Each card has a **"Submit to Marketplace" button.**

3. Investor Journey: Frontend

This is the "Scouting Dashboard" (your paid B2B product).

- **Trigger:** A user with the "investor" role logs in.
- **Feature:** A **read-only** gallery of all *public* innovator projects.
- **UI Components:**
  1. **Filter Bar:** `Filter by SDG:` (Checkboxes for `SDG 2` , `SDG 6` , etc.).
  2. **Project Gallery:** A list of all projects where `isPublic == true` .
  3. **Project Card:** Shows:
     - `Problem:` The original unsolved problem.
     - `Innovator's Plan:` The innovator's brief solution plan.
     - `Innovator:` The innovator's name (Sharvari Kinge).
     - `Tags:` `SDG 2` , `Agri-Tech` .

4. Backend & AI Pipeline (MVP Version)

This is the core logic.

A. Data Curation (Offline Task)

- **Who:** Rajshree (Data Curator).
- **Process:**
  1. Finds a high-quality paper (e.g., from IPCC).
  2. **CRITICAL:** Tags the paper with relevant **SDGs** (e.g., `["SDG 2", "SDG 6"]`).
  3. Uploads the paper to the folder that feeds the Vector Database.
- **Vector Database (Pinecone/Weaviate):** The database stores the text chunks *and* the metadata (`source_title` , `sdg_tags`).

B. AI Pipeline (When Innovator clicks "Discover")

- **Step 1. Get Request:** Receives the form data (Profession, Domain, Goal).
- **Step 2. Static Prompt Engineering:**
    - **NO "AI Enhancer" for MVP.** This is too complex.
    - Use a **static, well-crafted prompt template.**
    - `TEMPLATE = "You are an expert climate analyst. A [Profession] is looking for problems in [Domain]. Search the database for *unsolved technical challenges* or *validated research gaps*. **DO NOT INVENT SOLUTIONS.** Return the top 3 problem statements, each with its exact source citation. Filter your search only for documents tagged with [SDG tags for that Domain]."`
- **Step 3. RAG Query:**
    - The prompt and user goal are used to query the Vector DB.
    - **CRITICAL:** The query is *filtered* by the SDG tags (e.g., search for "water" only in docs tagged `SDG 6`).
- **Step 4. LLM Generation:**
    - **NO "Two LLM Validation" for MVP.** This is unnecessary. The *curated database* is your validation.
    - The prompt + retrieved chunks are sent to **one LLM call**.
    - The LLM synthesizes the text into the 3 structured "Challenge" objects.
- **Step 5. Send to Frontend:** The 3 challenges are sent to the results page.

5. Project Submission Logic

This is the "Marketplace" feature.

- **Trigger 1:** Innovator clicks "Start Project" on a challenge.
- **Action 1:** Creates a **new document** in your **Firestore** `projects` **collection**.
    - `projectId` (auto-generated)
    - `innovatorId` (from user)
    - `originalProblem` (from AI)
    - `originalSource` (from AI)
    - `sdgTags` (from AI)
    - `innovatorSolutionPlan` (A simple text box: "My plan is to...")
    - `architectureDiagramUrl` (Optional: PDF/Image upload URL)
    - `isPublic` (`false` by default)
- **Trigger 2:** Innovator goes to "My Impact" page and clicks "Submit to Marketplace" on a project.
- **Action 2:** The backend updates that project's document: `isPublic = true`.
- **Result:** The project instantly and automatically appears on the Investor Dashboard.